



OBJECT ORIENTED SOFTWARE ENGINEERING

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Special constructs of Activity Models

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Special Constructs of Activity Models

Activity diagram have additional notation that is useful for large and complex applications.

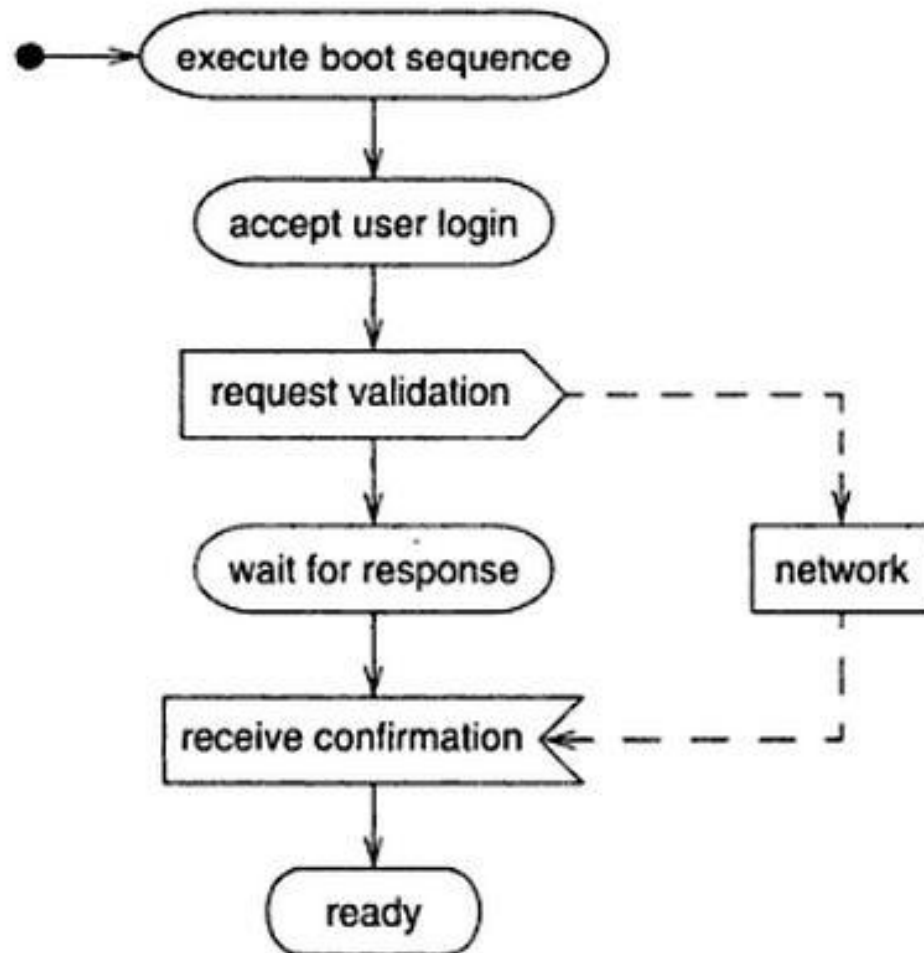
1. Sending and receiving signals:

Consider a work station that is turned on. It goes through a boot sequence and then requests the user login. After entry of name and password, the workstation queries the network to validate the user.

Upon validation, the work station then finishes the startup process.

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Special Constructs for Activity Models...



Here UML shows,

- sending of signal as a **convex pentagon**.
- Receiving of signal as **concave pentagon**

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Sending and Receiving signal construct

Activity diagram with signal construct

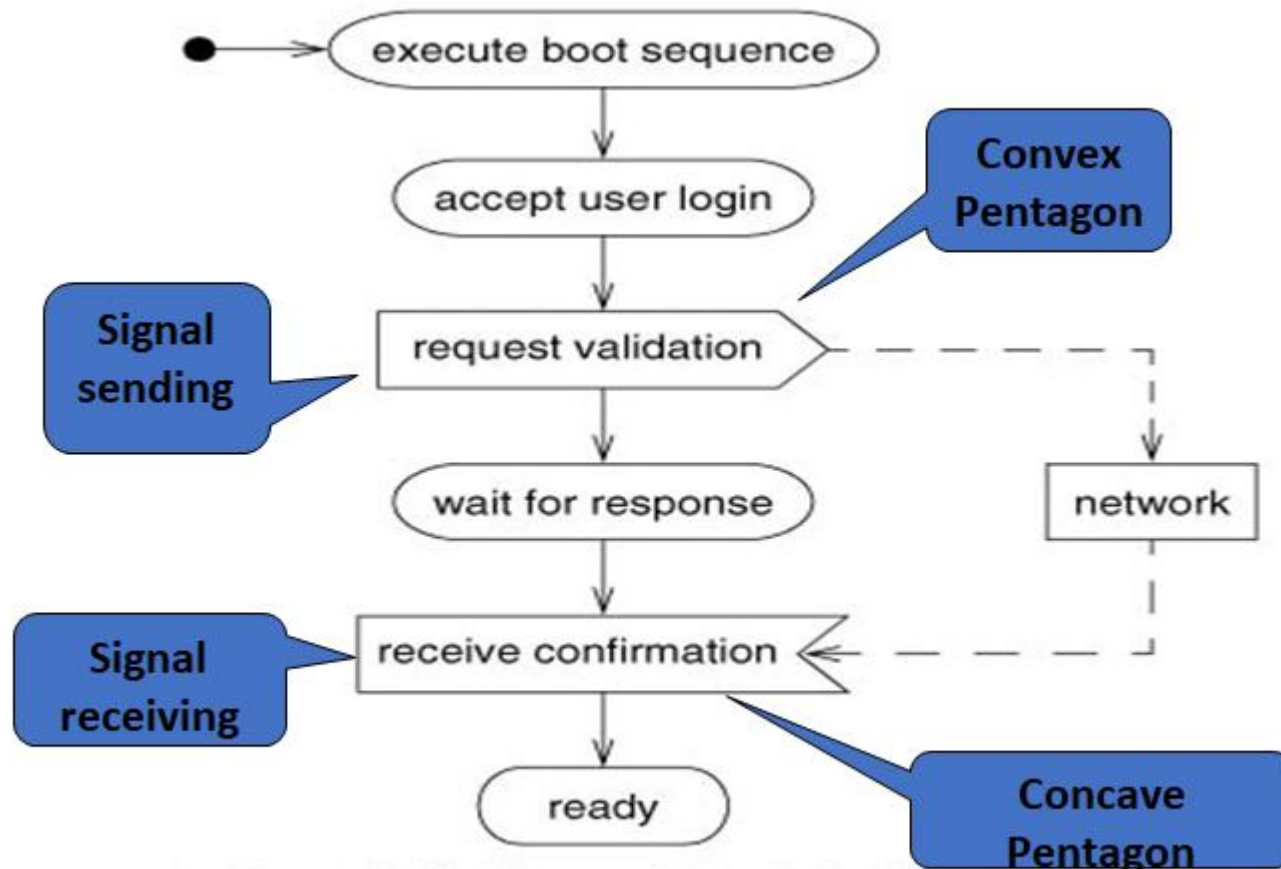


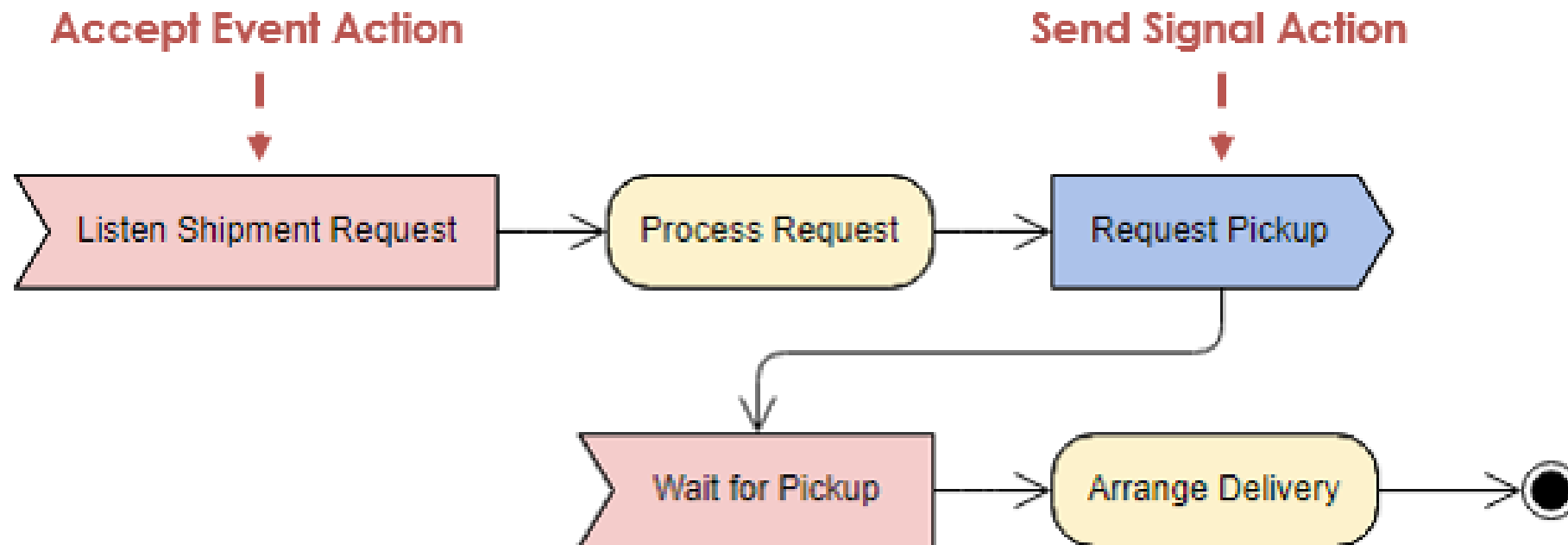
Figure Activity diagram with signals.

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Special Constructs for Activity Models...

Another Example: Here UML shows,

- sending of signal as a **convex pentagon**.
- Receiving of signal as **concave pentagon**



2. Swim Lane:

- In a business model, it is often useful to know which human organization is responsible for an activity.
Eg:-sales, marketing, purchasing.
- When the design of the system is complete, the activity will be assigned to a person.
- But at a higher level it is sufficient to partition the activities among organizations.

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Special Constructs for Activity Models...

You can show such a **partition** with an activity diagram **by dividing** it into **columns and lines**.

Each column is called **swim lane** (by analogy to a swimming pool).

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Swim Lane

Lines across Swim lane boundaries indicate **interaction among different Modules or organizations**

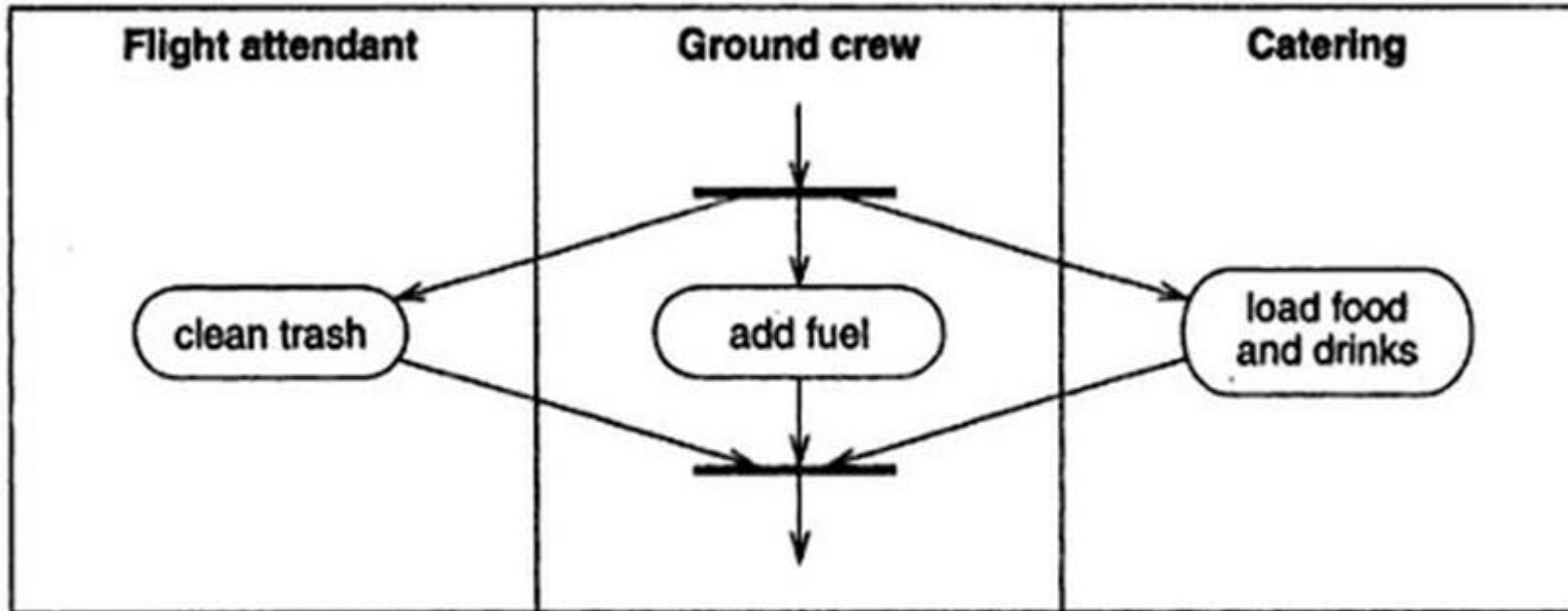


Figure Activity diagram with swimlanes. Swimlanes can show organizational responsibility for activities.

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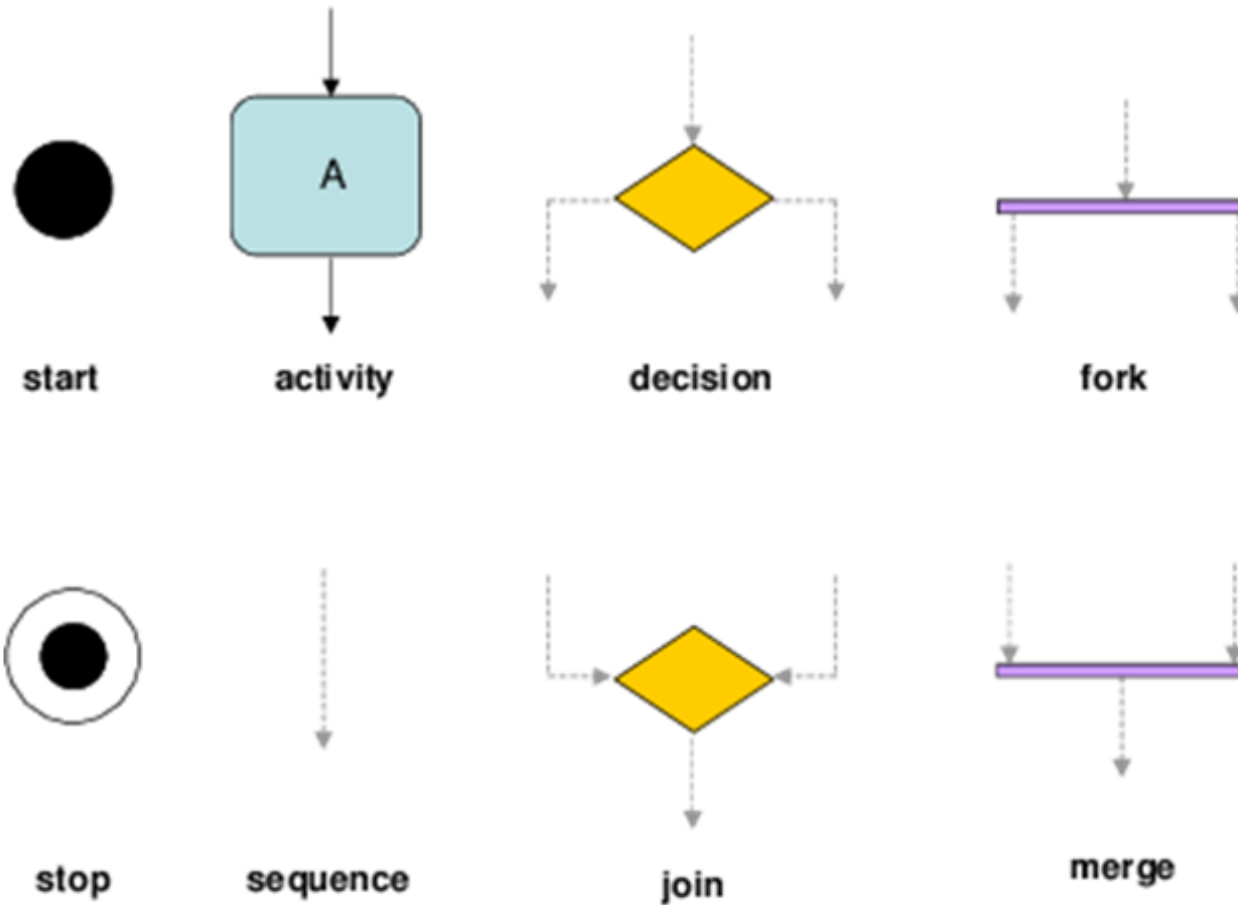
Swim Lane...

In the given example

- the flight attendants must clean the trash,
- the ground crew must add fuel and
- catering must load food and drink before a plane is serviced and ready for its next flight

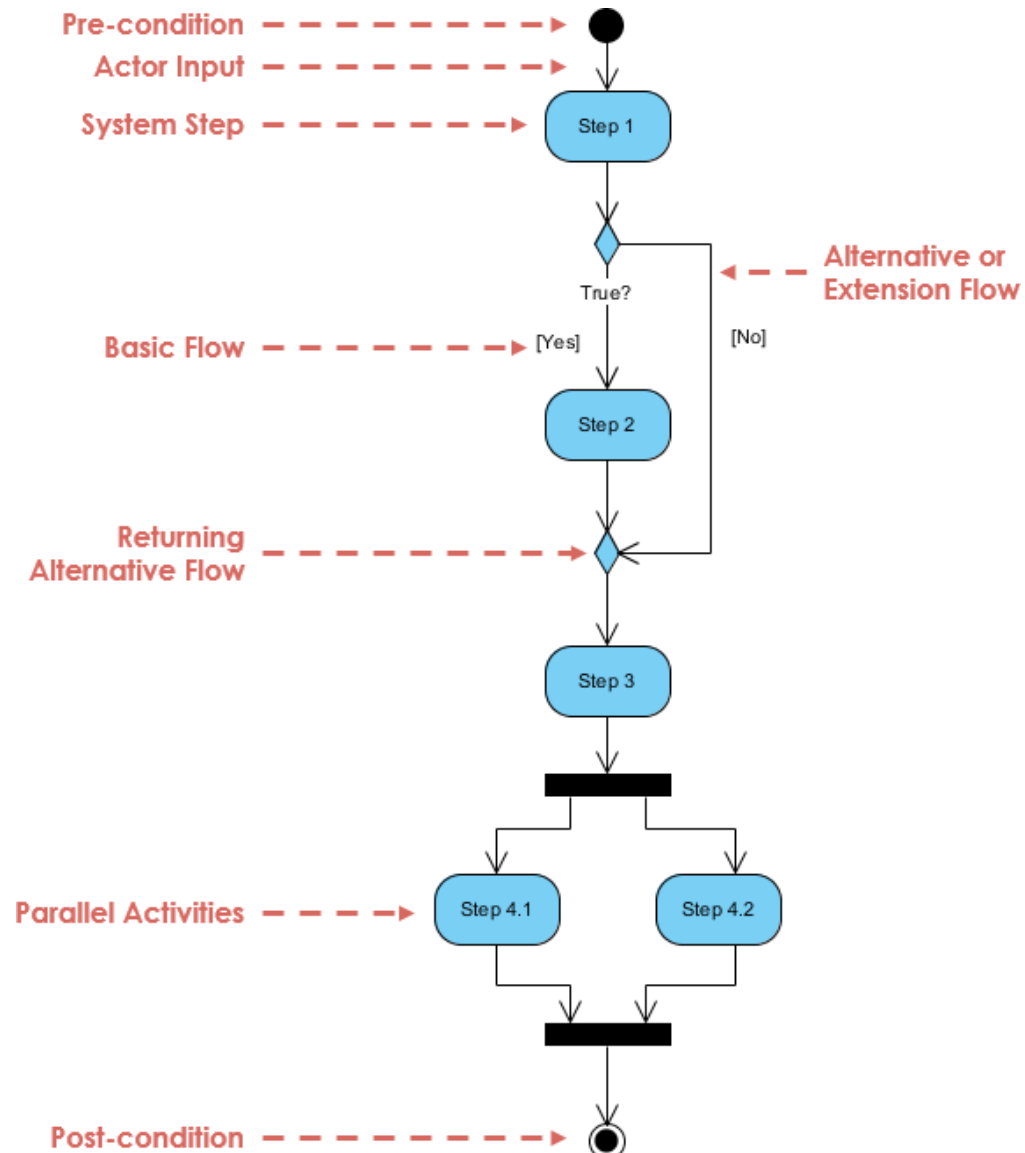
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Components used in Swim Lane Activity Diagram



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Components used in Swim Lane Activity Diagram...



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Swim Lane...

Consider an example

- **Ticket Vending machine**

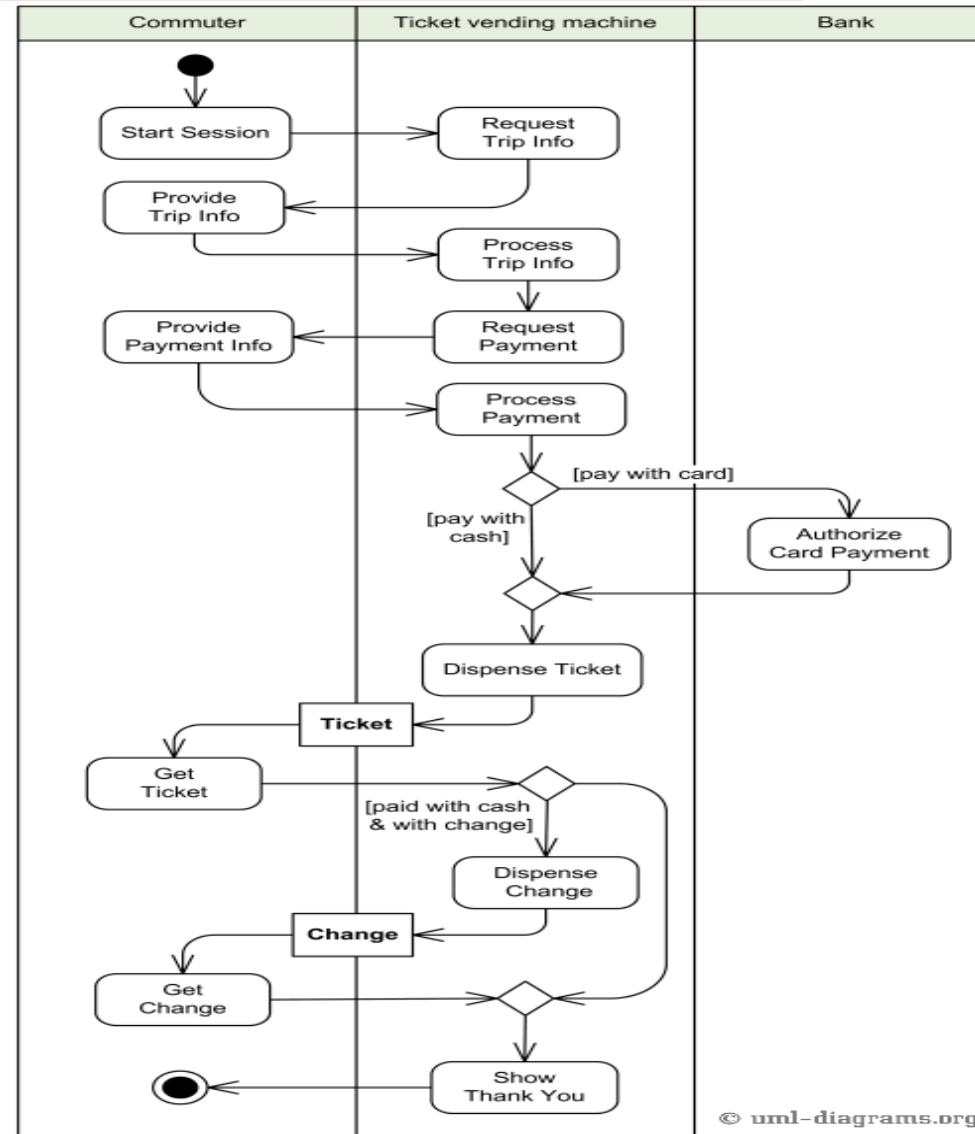
Description: Activity is started by Commuter actor who needs to buy a ticket. Ticket vending machine will request trip information from Commuter. Based on the info machine will calculate payment due and request payment options. After payment is complete, ticket is dispensed to the Commuter

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Swim Lane...

example

- Ticket Vending machine



Classroom Activity

Draw an Activity diagram for

.1. Restaurant Order Process:

Swimlanes: Customer, Waiter, Kitchen

Activities:

Customer: Choose meal, Place order, Receive food, Pay bill

Waiter: Take order, Send order to kitchen, Deliver food, Receive payment

Kitchen: Prepare food, Send food to waiter

2. Loan Application Process:

Swimlanes: Applicant, Bank, Credit Bureau

Activities:

Applicant: Fill out application, Submit documents

Bank: Verify information, Run credit check, Make loan decision, Disburse funds

Credit Bureau: Provide credit report



THANK YOU

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